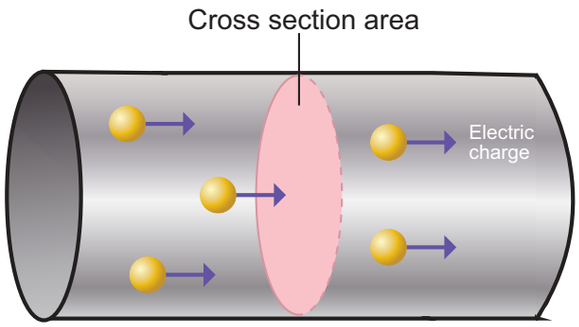
2.1. Electric Current

The flow of electric charges constitute an electric current. For an electrical appliance to work, electric current must flow through it. An electric current is measured by the amount of electric charge moving per unit time at any point in the circuit. The conventional symbol for current is 'I'.



Unit of Electric Current

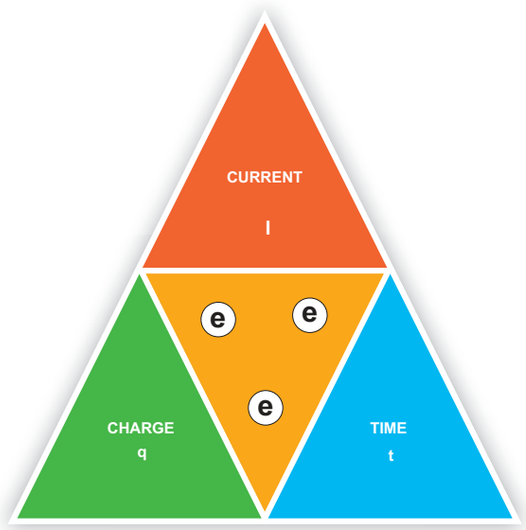
The SI unit for measuring an electric current is the **ampere**, which is the flow of electric charge across a surface at the rate of one coulomb per second.

$$I = \frac{q}{t}$$

Where $I \Rightarrow$ current (in Ampere - A)

$q \Rightarrow$ charge (in coulomb - c)

$t \Rightarrow$ time taken (in seconds - s)



Worked example 2.1

If 30 coulomb of electric charge flows through a wire in two minutes, calculate the current in the wire?

Solution

Given :

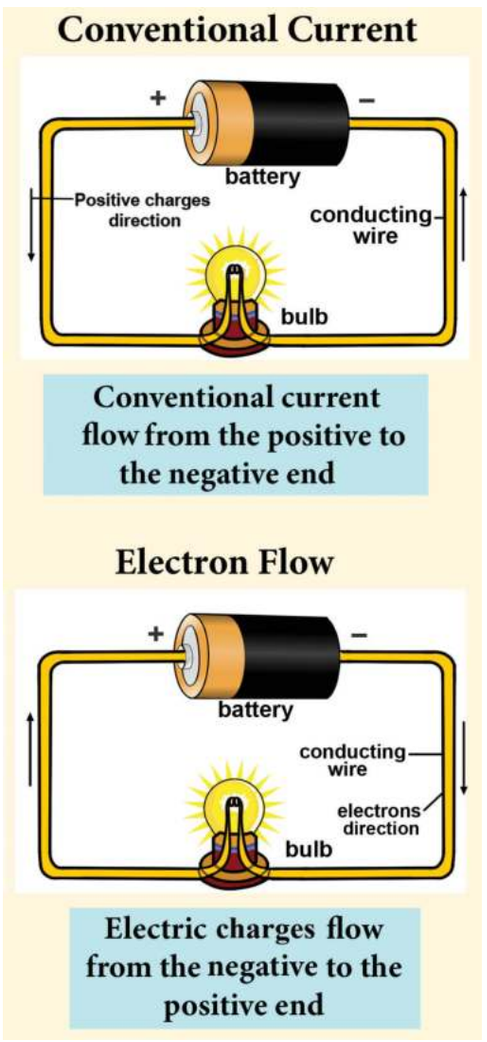
Charge (q) = 30 coulomb

Time (t) = 2 min x 60s

= 120 s

Current I = q/t = 30C/120s = 0.25 A

2.1.1. Conventional Current and Electron Flow



Before the discovery of electrons, scientists believed that an electric current consisted of moving positive charges.

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